

7.10 Springs

1. A maintenance manual tells you to reject any spring that has pitting on its surface. What is the main safety reason for this limit? /1
 - ☐ Pitting from overheating makes the spring too stiff during operation.
 - ☒ **Pitting from corrosion can cause cracks and lead to spring failure.**
 - ☐ Pitting from impact causes a small change in free length but no strength loss.

2. During a landing gear overhaul, a compression spring has the correct free length. The maintenance data also says that when the spring is compressed to 40 mm it must give a specified load. What additional check should you carry out on this spring? /1
 - ☒ **Put the spring in a spring tester, compress it to 40 mm, and make sure that the force is within the specified limits**
 - ☐ Measure the free length again and record the new value in the inspection report
 - ☐ Measure the coil diameter at several points and compare it with the nominal diameter in the manual

3. During an inspection, you find several small pits on the surface of a landing gear spring, but there is no bluing or distortion. What is the most likely cause of this defect? /1
 - ☐ Impact damage from a hard landing
 - ☐ Overheating during operation of the landing gear
 - ☒ **Corrosion of the spring material**

4. Using a spring tester, you compress a spring to 35 mm as required /1
by the maintenance manual. The tester shows the load and you
compare it with the specified range of 120 to 130 N. What are you
mainly checking with this test?

- ☐ That the spring returns exactly to its original free length after being compressed
- ☒ **That the force at 35 mm compression is within the specified load tolerance**
- ☐ That the spring surface does not show cracks or corrosion after compression